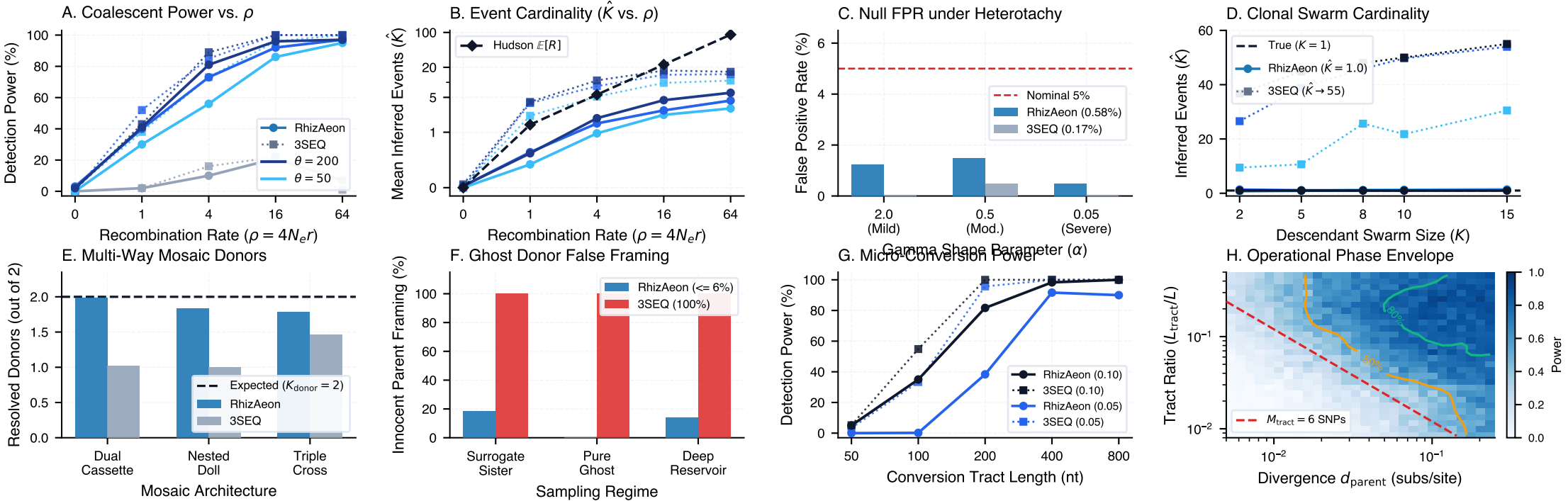

A. Coalescent Power vs. ρ
Detection Power (%)
Recombination Rate (ρ = 4N_e r)
RhizAeon
3SEQ
θ = 200
θ = 50
B. Event Cardinality (K̂ vs. ρ)
Mean Inferred Events (K̂)
Hudson 𝔼[R]
Recombination Rate (ρ = 4N_e r)
C. Null FPR under Heterotachy
False Positive Rate (%)
Nominal 5%
RhizAeon (0.58%)
3SEQ (0.17%)
2.0 (Mild)
0.5 (Mod.)
0.05 (Severe)
Gamma Shape Parameter (α)
D. Clonal Swarm Cardinality
Inferred Events (K̂)
True (K = 1)
RhizAeon (K̂ = 1.0)
3SEQ (K̂ → 55)
Descendant Swarm Size (K)
E. Multi-Way Mosaic Donors
Resolved Donors (out of 2)
Expected (K_donor = 2)
RhizAeon
3SEQ
Dual Cassette
Nested Doll
Triple Cross
Mosaic Architecture
F. Ghost Donor False Framing
Innocent Parent Framing (%)
RhizAeon (<= 6%)
3SEQ (100%)
Surrogate Sister
Pure Ghost
Deep Reservoir
Sampling Regime
G. Micro-Conversion Power
Detection Power (%)
RhizAeon (0.10)
3SEQ (0.10)
RhizAeon (0.05)
3SEQ (0.05)
Conversion Tract Length (nt)
H. Operational Phase Envelope
Tract Ratio (L_tract/L)
80%
50%
M_tract = 6 SNPs
Power
Divergence d_parent (subs/site)